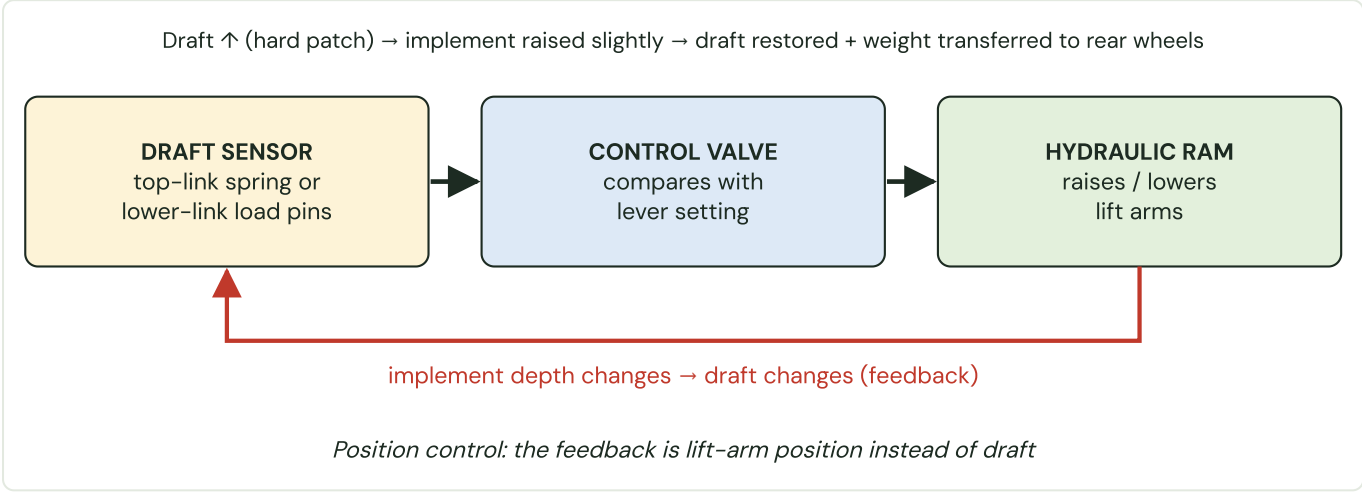
UNIT 4 · TOPIC 10

Position, Draft and Mixed Control

Syllabus topic: Hydraulic controls for tractors; automatic position and draft control



Automatic draft-control loop (redrawn from the source figure)

Control	Feedback / what is held	Best for
POSITION	Lift-arm position follows the quadrant lever — implement held at a SET HEIGHT/DEPTH regardless of draft (feedback = linkage position)	Uniform fields; raised transport; PTO implements (rotavator depth via gauge wheels); levelling boards
DRAFT	Senses DRAFT and raises/lowers the implement automatically to HOLD the set draft — depth shallows slightly in hard spots, deepens in soft	Tillage in varying soil; MB ploughs without depth wheels
MIXED (intermix)	Blends position and draft signals	Stability in variable soils

Draft sensing

- Light tractors: a **top-link compression spring**.
- Heavier and electronic (EHC) tractors: **lower-link sensing / load pins**.

Benefits of draft control

- Near-constant tractor load (no stalling).
- Automatic WEIGHT TRANSFER to the rear wheels in tough going (less slip).
- No depth wheels needed for MB ploughs.

Settings and ratings

- Response/drop-rate valves adjust sensitivity; over-sensitive response → HUNTING (rhythmic raise-lower) of the linkage.
- Lift capacity is rated at the hitch points and at **610 mm** behind them (standard test).

Practice MCQs

Q157. In POSITION control, feedback to the valve comes from:

(a) Draft spring
(b) Linkage/lift-arm position
(c) Wheel slip
(d) Fuel rack

Q158. In DRAFT control the quantity held constant is:

(a) Depth
(b) Height
(c) Implement draft (pull)
(d) Speed

Q159. Draft is sensed on small tractors by:

(a) Radar
(b) Top-link compression spring
(c) Ammeter
(d) Airflow

Q160. On modern/heavier tractors draft sensing is via:

(a) Lower-link load pins (electronic EHC)
(b) Rope
(c) Vacuum
(d) Belt

Q161. Under draft control in a hard soil patch, the implement is automatically:

(a) Deepened
(b) Raised slightly
(c) Stopped
(d) Widened

Q162. A key traction benefit of draft control is:

(a) Less fuel tank use
(b) Automatic weight transfer to rear wheels reducing slip
(c) Wider tyres
(d) Cooler engine

Q163. Rotavator working depth is normally managed with:

(a) Draft control
(b) Position control (+ depth/skid wheels)
(c) No control
(d) Brakes

Q164. MIXED control blends:

(a) Oil and water
(b) Position and draft signals
(c) Petrol and diesel
(d) Front and rear brakes

Q165. Hydraulic lift capacity is rated at the hitch points and at:

(a) 310 mm
(b) 610 mm behind them
(c) 1 m ahead
(d) The seat

Q174. Over-sensitive draft response causes:

(a) Smooth work
(b) Hunting (rhythmic raise-lower) of the linkage
(c) Stall
(d) Slip

Answers

Q157 — (b) Linkage/lift-arm position — *Position control = geometry feedback, constant HEIGHT.*

Q158 — (c) Implement draft (pull) — *Draft control = force feedback, constant PULL (auto depth + weight transfer).*

Q159 — (b) Top-link compression spring

Q160 — (a) Lower-link load pins (electronic EHC)

Q161 — (b) Raised slightly

Q162 — (b) Automatic weight transfer to rear wheels reducing slip

Q163 — (b) Position control (+ depth/skid wheels)

Q164 — (b) Position and draft signals

Q165 — (b) 610 mm behind them

Q174 — (b) Hunting (rhythmic raise-lower) of the linkage

Sources: Liljedahl et al. — Tractors and their Power Units; Kepner et al. (hitching); IS 4468/ISO 730; Jagdishwar Sahay; NPTEL lectures.